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Topological aspects of photonic time crystals: supplementary material

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This document provides supplementary information to "Topological aspects of photonic time crystals," https://doi.org/10.1364/OPTICA.5.001390. Here we give the mathematical details of the topology of the photonic time crystal and additional simulations.

Calculation of band structure and Zak phases.
In Fig. S1 we illustrate the photonic time-crystal (PTC). It is a binary PTC with two repeating segments. In the first time-segment $\epsilon(t) = \epsilon_1$ for a duration of t_1 seconds, followed by a second time-segment in which $\epsilon(t) = \epsilon_2$ for $t_2 = T - t_1$ seconds (Fig. S1). For simplicity, the field is polarized in the x direction, and propagates in the z direction.

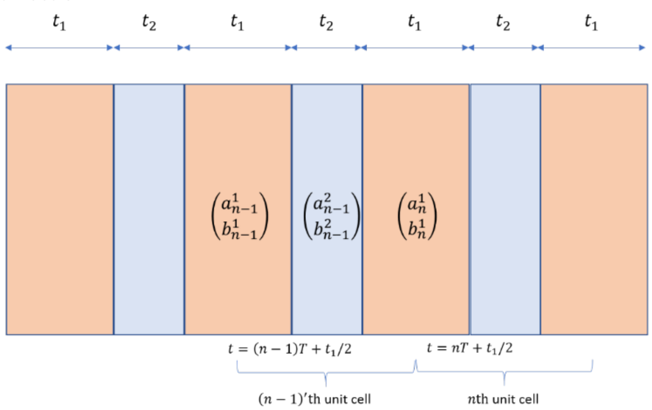

Fig. S1. The time lattice. Each color corresponds to a different refractive index, and the horizontal direction is time.

We approach the problem with the transfer matrix formalism [1]. The continuity conditions on a temporal boundary at $t = t_0$ between two time segments are given by

$$\boldsymbol{D}(t = t_0^+, z) = \boldsymbol{D}(t = t_0^-, z), \qquad \text{(S1)}$$

$$\boldsymbol{B}(t = t_0^+, z) = \boldsymbol{B}(t = t_0^-, z), \qquad \text{(S2)}$$

where $\boldsymbol{B}$ is the magnetic field and $\boldsymbol{D}$ is the displacement field. We fix the time axis such that the system will be symmetric to time inversion: $t \to -t$. $t = 0$ is set to be at the middle of a segment with duration t_1 (See Fig. S1). The time inversion symmetry is required for defining the topological invariants. The momentum k of a wave is conserved due to homogeneity of space, and during each time-segment the light has a dispersion relation:

$$\omega_\alpha = kc/n_\alpha, \qquad (S3)$$

where $\alpha = 1{,}2$ and n_α is the refractive index of segment α. Under these constraints the electric displacement of a wave with momentum k is given by

$$D_x(t,z) = \left[a_n^{(\alpha)} e^{-i\omega_\alpha \cdot \left(t + \frac{t_\alpha}{2} - nT\right)} + b_n^{(\alpha)} e^{i\omega_\alpha \cdot \left(t + \frac{t_\alpha}{2} - nT\right)}\right] e^{ik_z z}, (S4)$$

where $a_n^{(\alpha)}, b_n^{(\alpha)}$ are constant coefficients for each n, α. The magnetic field can be obtained through Maxwell's equations in each segment (away from the boundaries):

$$\nabla \times \boldsymbol{B} = \frac{1}{\mu_0} \frac{\partial \boldsymbol{D}}{\partial t}, \qquad (S5)$$

$$ik_z B_y = i\frac{\omega_\alpha}{\mu_0}\left[-a_n^{(\alpha)} e^{i\omega_\alpha \cdot \left(t + \frac{t_\alpha}{2} - nT\right)} + a_n^{(\alpha)} e^{-i\omega_\alpha \cdot \left(t + \frac{t_\alpha}{2} - nT\right)}\right], (S6)$$

where μ_0 is the magnetic permeability. At this point, it will be convenient to use the notations

$$a_n^{(1)} \equiv a_n, \qquad b_n^{(1)} \equiv b_n, \qquad a_n^{(2)} \equiv c_n, \qquad b_n^{(2)} \equiv d_n, \qquad (S7)$$

Substituting Eqs. (S5) and (S6) into the continuity conditions yields the transfer matrices for a_n, b_n, c_n, d_n:

$$\begin{pmatrix} a_{n-1} \\ b_{n-1} \end{pmatrix} = \begin{pmatrix} X_1 & Y_1 \\ Z_1 & W_1 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}, \qquad \text{(S8)}$$

$$\begin{pmatrix} c_{n-1} \\ d_{n-1} \end{pmatrix} = \begin{pmatrix} X_2 & Y_2 \\ Z_2 & W_2 \end{pmatrix} \begin{pmatrix} c_n \\ d_n \end{pmatrix}, \qquad \text{(S9)}$$

where $X_\alpha = e^{i\omega_\alpha t_a}\left[\cos(\omega_{\bar{\alpha}} t_{\bar{\alpha}}) + \frac{1}{2} i \left(\frac{\omega_2}{\omega_1} + \frac{\omega_1}{\omega_2}\right) \sin \omega_{\bar{\alpha}} t_{\bar{\alpha}}\right]$,

$Y_\alpha = \frac{i}{2} e^{-i\omega_\alpha (T(\alpha) + t_a)} \left(\frac{\omega_{\bar{\alpha}}}{\omega_\alpha} - \frac{\omega_\alpha}{\omega_{\bar{\alpha}}}\right) \sin \omega_{\bar{\alpha}} t_{\bar{\alpha}}, Z_\alpha = Y_\alpha^*, W_\alpha = X_\alpha^*$,

$T(\alpha) = \begin{cases} 0 & \alpha = 1 \\ T & \alpha = 2 \end{cases}$ and

$\bar{\alpha} = \begin{cases} 2 & \alpha = 1 \\ 1 & \alpha = 2 \end{cases}$.

Equations (S8) and (S9) give the fields D_x and B_y for any initial conditions. However, the topological properties are defined for the infinite system. Since $\epsilon(t)$ is periodic in time, according to the Floquet-Bloch theorem, the displacement field is of the form

$$D_x(t,z) = D_\Omega(t,z) e^{-i\Omega t}, \qquad \text{(S10)}$$

where $D_\Omega(t,z)$ is periodic with period T. Applying condition (S10) on Eqs. (S8) and (S9) gives the Floquet dispersion:

$$\Omega(k_z) = \frac{1}{T} \cos^{-1}\left(\frac{1}{2}(W + X)\right), \qquad \text{(S11)}$$

where we used the fact that $W+X=W_\alpha+X_\alpha$, and the eigenmodes

$$\begin{pmatrix} a_0^\alpha \\ b_0^\alpha \end{pmatrix} = \begin{pmatrix} Y_\alpha \\ e^{i\Omega T}-X_\alpha \end{pmatrix}. \qquad \text{(S12)}$$

Thus, in the infinite lattice the modes of the displacement field are given by

$$D_\Omega(t,z) = \left[a_0^{(\alpha)} e^{-i\omega_\alpha\cdot\left(t+\frac{t_\alpha}{2}-nT\right)} + b_0^{(\alpha)} e^{i\omega_\alpha\cdot\left(t+\frac{t_\alpha}{2}-nT\right)}\right] e^{-i\Omega nT} e^{ik_z z}. \quad \text{(S13)}$$

The bands are where Ω takes real values, and the gaps are where Ω takes imaginary values.

Next, we define the Zak phase for the displacement field D_x. The Zak phase is given by

$$\theta_m^{Zak} = i\int_{-\frac{\pi}{T}}^{\frac{\pi}{T}} d\Omega \int_{temporal\ period} dt\, D^*_{m,\Omega}(t,z)\,\partial_\Omega D_{m,\Omega}(t,z)\,, \quad \text{(S14)}$$

where m is the band number and $D_{m,\Omega}$ is D_Ω of band m. We wish to prove that the Zak-phase is quantized to 0 or π. Since $D_{m,\Omega}$ are normalized modes composed of orthogonal elements (with respect to temporal integration) we can rewrite them as

$$D_{m,\Omega} = \begin{pmatrix} Y_\alpha \\ e^{i\Omega T}-X_\alpha \end{pmatrix} \equiv \begin{pmatrix} 1 \\ e^{i\phi_\alpha} \end{pmatrix}, \qquad \text{(S15)}$$

where we used Eq. (S12). Using this notation, the integrand of Eq. (S14) is $\partial_\Omega\phi_a$. $\partial_\Omega\phi_a$ is given by

$$\partial_\Omega\phi_a = \frac{i\mathrm{T}e^{i\Omega T}}{e^{i\Omega T}-X_\alpha}. \qquad \text{(S16)}$$

If the denominator of Eq. (S16) is different than zero, the θ_m^{Zak} is zero since the function has no poles on the integration path. However, for bands in which $e^{i\Omega T}-X_\alpha$ equals zero, $\partial_\Omega\phi_a$ has a pole; therefore, $\theta_m^{Zak}=\pi$.

Next, we will show that the phase between the two emerging pulses of the PTC is calculated through the Zak phase. Consider an incoming plane wave $D_{in}=D_0e^{-i\omega t}$. The reflected wave from the PTC is $D_r=A_rD_0e^{i\omega t}$ and the transmitted wave is $D_t=A_tD_0e^{-i\omega t}$. We take k to be inside bandgap number s. A moment before the PTC ends, at time $t=nT^-$, the electric displacement field is described by the Floquet-Bloch state in Eq. (S13). To find the expression for the phase between D_t and D_r we use the continuity of D_x and B_y:

$$\frac{D_x(nT^+)}{B_y(nT^+)} = \frac{D_x(nT^-)}{B_y(nT^-)}. \qquad \text{(S17)}$$

Substituting the Floquet-Bloch modes on the left hand side of Eq. (S17) and the reflection and transmission coefficients on the right hand side gives, after some algebraic manipulation,

$$\frac{1+\frac{A_r}{A_t}}{1-\frac{A_r}{A_t}} = \frac{n_{\bar\alpha}}{n_\alpha}\frac{B_\alpha e^{-i\omega_\alpha t^*} + \left(e^{i\Omega T}-A_\alpha\right)e^{i\omega_\alpha t^*}}{B_\alpha e^{-i\omega_\alpha t^*} - \left(e^{i\Omega T}-A_\alpha\right)e^{i\omega_\alpha t^*}}\,, \qquad \text{(S18)}$$

where $t^*=t_{end}+t_1/2$ and t_{end} is the duration of the lattice. We define the phase ϕ_s to be the phase between the refracted wave and the transmitted wave remaining after the PTC ends. Since A_r and A_t are equal in absolute value we have

$$\frac{A_r}{A_t} = e^{i\phi_s}. \qquad \text{(S19)}$$

From Eqs. (S18) and (S19) it is possible to calculate the phase [2] and show that the phase sign of ϕ_s is related to the Zak phase according to

$$\mathrm{sgn}(\phi_s) = \delta(-1)^s(-1)^l\exp\left(i\sum_{m=1}^{s-1}\theta_m{}^{Zak}\right), \qquad \text{(S20)}$$

where s is the gap number (lowest gap number is 1), $\delta=\mathrm{sgn}(1-\epsilon_a\mu_b/\epsilon_b\mu_a)$, and l is the number of metallic band crossings below gap s.

Discussion on the Topological Zak phase for 1D smooth systems.

In the previous section, we calculated rigorously the quantization of the Zak-phase for a particular ideal PTC, where the electric permittivity varies abruptly in a periodic manner. As in most topological systems, the Zak phase of the ideal PTC has physical implications on many other PTCs which conserve the time-reversal and periodicity symmetries. In this section we show that when the electric permittivity varies smoothly (rather than abruptly, as for the ideal PTC) the system still displays the same topological properties.

Consider the wave equation for a homogenous time-varying material (assuming instantaneous response of the polarization):

$$\nabla^2\boldsymbol{E} = \epsilon(t)\mu\frac{\partial^2\boldsymbol{E}}{\partial t^2} + 2\mu\dot\epsilon(t)\frac{\partial\boldsymbol{E}}{\partial t} + \mu\ddot\epsilon(t)\boldsymbol{E}, \qquad \text{(S21)}$$

where $\epsilon(t)=\epsilon(t+T)$, μ is constant and $\boldsymbol{E}$ is the electric field. Since $\epsilon(t)$ is periodic, Floquet theorem applies so that $\boldsymbol{E}=\boldsymbol{E}_\Omega e^{i\Omega t}$, where $\boldsymbol{E}_\Omega$ is periodic in time with period T. We write Eq .(S19) with the Floquet wavefunction:

$$k_{z,m}(\Omega)^2\boldsymbol{E}_{m,\Omega} = \left[\epsilon(t)\mu\left(\frac{\partial}{\partial t}+i\Omega\right)^2 + 2\mu\dot\epsilon(t)\left(\frac{\partial}{\partial t}+i\Omega\right) + \mu\ddot\epsilon(t)\right]\boldsymbol{E}_{m,\Omega}, \qquad \text{(S22)}$$

where m is the band number, $k_z(\Omega)$ is periodic in Ω with a period $\frac{2\pi}{T}$, the equation has band gaps [3], and $k_{z,m}(\Omega)$ means $k_z(\Omega)$ for band m. The Floquet modes of Eq. (S22) obey time reversal symmetry (since $\dot\epsilon(t)=-\dot\epsilon(-t)$), and therefore display the symmetries required for a quantized Zak phase.

Studying the topological properties of an ideal PTC (where the electric permittivity varies periodically in an abrupt manner) and the more physical case of permittivity that varies periodically in a smooth manner, raises a natural question about the topological properties of a medium whose $\epsilon(t)$ varies periodically in an arbitrary manner. However, the rigorous calculation of the quantized Zak phases for the arbitrary $\epsilon(t)$ would involve using a biorthogonal basis as in [5], and is beyond the scope of this work. Furthermore, physically, the electric susceptibility memory implies a more general relation: $D(t)=\int_0^\infty f(\tau,t)E(t-\tau)\,d\tau$ where $f(\tau,t)$ is a time varying linear response function, which adds further complications [4]. However, at least for the specific example of the periodic smoothened $\epsilon(t)$ described in Fig. 5(c) in the main text, we find – in simulations [according to Eq. (S21)] – that the topological properties are conserved. The results are shown in Figs. S2(a)-S2(d), where the phase between the forward and backward propagating waves is plotted for the first four band gaps.

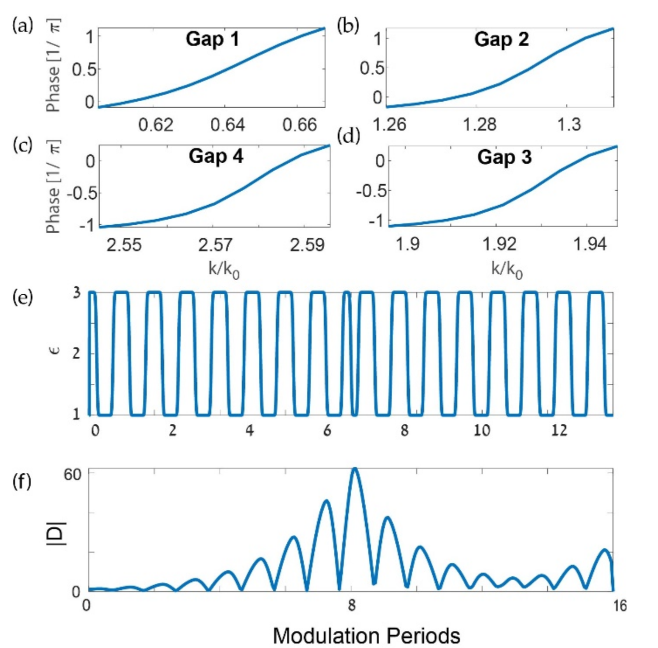

Figure S2: (a)-(d) Phase difference ϕ between the forward- and backward-propagating Floquet modes of the first four gaps of the smooth PTC from Fig. 5(c) in the main text, obtained from FDTD simulations. (e) Two smooth PTCs with different Zak phases cascaded at $t = 6T$. (f) Displacement field's amplitude of the pulse propagating in the PTC plotted in (e) displaying a temporal edge-state.

While the PTC and therefore the band gaps are now different, the phase signs of the first four band gaps remain the same as in the binary PTC. Gaps number 5 and 6 no longer exist or are vanishingly small, and therefore the phase could not be obtained for them.

Finally, we show that edge states also exist in the system after the binary crystal is 'smoothened'. Figures S2 (e)-(f) show the electromagnetic displacement field's amplitude in a temporal system comprising of two sequential PTCs. The two PTCs have the same ϵ values but in reverse order. Just like in the temporal system described in the main text, which yielded a temporal edge state, here we also effectively have two sequential PTCs with the same band gaps but different Zak phases. Therefore, light inside of the band gap experiences an edge state at their temporal interface. The exact $\epsilon(t)$ of this system can be seen in Fig. S2(e), and the field's amplitude in the system in Fig. S2(f). Just like in the binary case, we observe a clear edge state, centered at the temporal interface between the two PTCs.

References

1. M. C. Rechtsman, J. M. Zeuner, Y. Plotnik, Y. Lumer, D. Podolsky, F. Dreisow, S. Nolte, M. Segev, and A. Szameit, "Photonic Floquet topological insulators," Nature **496**, 196–200 (2013).
2. M. Xiao, Z. Q. Zhang, and C. T. Chan, "Surface impedance and bulk band geometric phases in one-dimensional systems," Phys. Rev. X **4**, 21017 (2014).
3. J. R. Zurita-Sánchez, P. Halevi, and J. C. Cervantes-González, "Reflection and transmission of a wave incident on a slab with a time-periodic dielectric function ϵ(t)," Phys. Rev. A **79,** 53821 (2009).
4. L. D. Landau, E. M. Lifshitz, and A. L. King, *Electrodynamics of Continuous Media, Course of Theoretical Physics* (Butterworth, 1961), Vol. 29.
5. S. Lieu, "Topological phases in the non-Hermitian Su-Schrieffer-Heeger model," Phys. Rev. B **97,** 45106 (2018).